

A Consistency-Gated Quaternion Error-State Kalman Filter for Multi-Sensor Inertial Navigation

Irfan Saric

December 2025

Abstract

We describe the design, implementation, and verification of an error-state Kalman filter (ESKF) that fuses a strapdown inertial measurement unit (IMU) with up to nine aiding modalities—GPS position and Doppler velocity, a barometric altimeter, a magnetometer, a downward laser altimeter (LiDAR), ultra-wideband (UWB) radio ranging, optical flow, a Doppler velocity log (DVL), and a direct attitude fix—to estimate the pose of an aerial vehicle. The filter follows the multiplicative, local error-state formulation of Solà [2] and Trawny and Roumeliotis [3]: a nominal state carries the full estimate and is integrated directly from the IMU, while a minimal 15-dimensional error state, reset to zero after every correction, carries the covariance. Each of the ten measurement models is given with its exact error-state Jacobian, and every Jacobian is validated against a finite difference. The central claim is not merely that the filter tracks well but that its *reported uncertainty is correct*: we verify this with the normalized estimation error squared (NEES) consistency test of Bar-Shalom et al. [5], run as a Monte-Carlo gate over many simulated flights and shown to hold throughout each flight, not only in the mean. On the full nine-sensor fusion the filter is statistically consistent (position NEES 2.85 against an expected 3.0; full-state NEES 16.2 against 15.0) at 0.05 m position RMSE, while remaining faithful to inertial-navigation error theory: with all aiding removed the position error grows without bound, and it is recovered to sub-metre accuracy by any single position-fixing modality, including GPS-denied UWB ranging. A recorded reference dataset and a replay tool accompany the filter, and the complete system—core, simulator, and a real-time 3D visualization—runs in the browser via WebAssembly and WebGL2.

1 Introduction

Aided inertial navigation estimates a vehicle’s position, velocity, and orientation by integrating high-rate inertial measurements and correcting the accumulated drift with lower-rate aiding sensors [6, 7, 8]. Orientation makes the problem subtle: a unit quaternion has four components but only three degrees of freedom, so a filter that estimates it directly must either enforce the unit-norm constraint or carry a covariance for a quantity that is not free to move in four directions. The *error-state* (or indirect, or multiplicative) Kalman filter resolves this by estimating a minimal three-parameter rotation error in the tangent space of the current attitude estimate [4, 2]. This paper documents such a filter, its ten measurement models, and—most importantly—the statistical test that proves its covariance honest.

The contribution is one of rigor and reproducibility rather than novelty of formulation. The filter is a faithful implementation of the standard local-error ESKF, but (i) each of the ten measurement Jacobians is finite-difference-checked; (ii) the whole estimation loop is gated on NEES consistency, verified both in aggregate and as a function of time; (iii) a recorded reference dataset and a replay harness make the results independently reproducible; and (iv) the entire system, from the dependency-free numerical core to a real-time three-dimensional visualization, runs

in the browser. Source, the dataset, and an interactive demonstration are available at <https://github.com/Zarathos94/ekf-sim> and <https://irfansaric.com/showcase/ekf-sim>.

2 Related Work and Datasets

The indirect Kalman filter for attitude dates to spacecraft navigation and is surveyed by Markley [4] and, in quaternion form, by Trawny and Roumeliotis [3] and Solà [2]; it is the attitude workhorse of aided-INS texts [6, 7]. For visual-inertial estimation the multi-state constraint Kalman filter (MSCKF) of Mourikis and Roumeliotis [9] keeps a sliding window of poses, and optimization-based systems such as VINS-Mono [10] and the factor-graph / incremental-smoothing approach of iSAM2 and GTSAM [12, 13] trade recursion for re-linearization. The invariant EKF of Barrau and Bonnabel [11] improves consistency by exploiting the group structure of the state; the filter here uses the more common local-error parameterization but demonstrates consistency empirically. Radio ranging with UWB for micro-aerial vehicles follows Mueller et al. [14], and metric optical flow follows Honegger et al. [15].

Public datasets are the standard proving ground for such estimators: EuRoC MAV [16] and TUM-VI [17] provide synchronized IMU, stereo imagery, and millimetre-accurate motion capture ground truth; UZH-FPV [18] targets aggressive first-person-view flight; and KITTI [19] provides automotive IMU, GNSS, and LiDAR. Because ground truth on such data is itself an estimate and the sensor suites differ from ours, we validate against an *analytic* trajectory—where truth is exact—and additionally ship a recorded, replayable dataset (Section 8) so that our specific results are reproducible bit-for-bit.

3 State Representation

The **nominal state** collects position $\mathbf{p} \in \mathbb{R}^3$, velocity $\mathbf{v} \in \mathbb{R}^3$, orientation as a Hamilton unit quaternion $\mathbf{q} \in \mathbb{S}^3$ (body to world), and the accelerometer and gyroscope biases $\mathbf{a}_b, \boldsymbol{\omega}_b \in \mathbb{R}^3$:

$$\mathbf{x} = (\mathbf{p}, \mathbf{v}, \mathbf{q}, \mathbf{a}_b, \boldsymbol{\omega}_b). \quad (1)$$

The **error state** is the 15-vector

$$\delta\mathbf{x} = (\delta\mathbf{p}, \delta\mathbf{v}, \delta\boldsymbol{\theta}, \delta\mathbf{a}_b, \delta\boldsymbol{\omega}_b) \in \mathbb{R}^{15}, \quad (2)$$

where the rotation error $\delta\boldsymbol{\theta}$ is defined *locally* (in the body frame) through the multiplicative relation

$$\mathbf{q}_{\text{true}} = \mathbf{q} \otimes \exp\left(\frac{1}{2}\delta\boldsymbol{\theta}\right), \quad (3)$$

with $\exp(\cdot)$ the quaternion exponential map (Appendix A) and $\otimes$ the Hamilton product. The nominal state holds the full, unconstrained estimate; the error state, kept near zero, holds a covariance $\mathbf{P} \in \mathbb{R}^{15 \times 15}$ whose orientation block is an ordinary 3×3 matrix. We write $\mathbf{R} = \mathbf{R}(\mathbf{q})$ for the rotation matrix and $[\cdot]_{\times} : \mathbb{R}^3 \rightarrow \mathfrak{so}(3)$ for the skew-symmetric operator, so that $[\mathbf{a}]_{\times} \mathbf{v} = \mathbf{a} \times \mathbf{v}$.

Figure 1 shows the overall structure, and Algorithm 1 the per-step loop.

4 Propagation

At each IMU sample (rate ≈ 200 Hz, interval Δt) with accelerometer reading $\mathbf{a}_m$ and gyroscope reading $\boldsymbol{\omega}_m$, define the bias-corrected specific force and body rate $\mathbf{a} = \mathbf{a}_m - \mathbf{a}_b$ and $\boldsymbol{\omega} = \boldsymbol{\omega}_m - \boldsymbol{\omega}_b$.

Nominal integration. With gravity $\mathbf{g} = (0, 0, -9.80665 \text{ m s}^{-2})$,

$$\mathbf{p} \leftarrow \mathbf{p} + \mathbf{v} \Delta t + \frac{1}{2}(\mathbf{R}\mathbf{a} + \mathbf{g}) \Delta t^2, \quad \mathbf{v} \leftarrow \mathbf{v} + (\mathbf{R}\mathbf{a} + \mathbf{g}) \Delta t, \quad \mathbf{q} \leftarrow \mathbf{q} \otimes \exp\left(\frac{1}{2}\boldsymbol{\omega} \Delta t\right), \quad (4)$$

with the quaternion renormalized after each step.

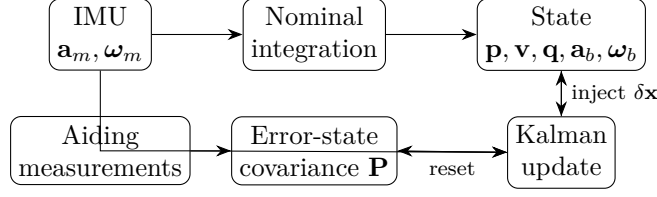


Figure 1: The error-state filter. The IMU drives the nominal state forward and inflates the error-state covariance; each aiding measurement produces a correction that is injected into the nominal state, after which the error state is reset to zero.

Algorithm 1 Error-state Kalman filter, one cycle

- 1: **Predict** (each IMU sample): integrate $\mathbf{p}, \mathbf{v}, \mathbf{q}$; set $\mathbf{P} \leftarrow \mathbf{F}\mathbf{P}\mathbf{F}^\top + \mathbf{Q}$
 - 2: **for all** available measurements $\mathbf{z}$ with model h , Jacobian $\mathbf{H}$, noise $\mathbf{R}$ **do**
 - 3: $\mathbf{S} \leftarrow \mathbf{H}\mathbf{P}\mathbf{H}^\top + \mathbf{R}$; $\mathbf{K} \leftarrow \mathbf{P}\mathbf{H}^\top\mathbf{S}^{-1}$; $\delta\mathbf{x} \leftarrow \mathbf{K}(\mathbf{z} - h(\mathbf{x}))$
 - 4: **Inject**: $\mathbf{x} \leftarrow \mathbf{x} \boxplus \delta\mathbf{x}$; reset $\delta\mathbf{x} \leftarrow \mathbf{0}$; $\mathbf{P} \leftarrow \mathbf{G}\mathbf{P}\mathbf{G}^\top$
 - 5: $\mathbf{P} \leftarrow (\mathbf{I} - \mathbf{K}\mathbf{H})\mathbf{P}(\mathbf{I} - \mathbf{K}\mathbf{H})^\top + \mathbf{K}\mathbf{R}\mathbf{K}^\top$ ▷ Joseph form
 - 6: **end for**
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Error-state transition. $\delta\mathbf{x} \leftarrow \mathbf{F}\delta\mathbf{x}$ with the block matrix

$$\mathbf{F} = \begin{bmatrix} \mathbf{I} & \mathbf{I}\Delta t & \mathbf{0} & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{I} & -\mathbf{R}[\mathbf{a}]_\times \Delta t & -\mathbf{R}\Delta t & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \exp(-[\boldsymbol{\omega}]_\times \Delta t) & \mathbf{0} & -\mathbf{I}\Delta t \\ \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{I} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{I} \end{bmatrix}, \quad (5)$$

in which the velocity–orientation coupling $-\mathbf{R}[\mathbf{a}]_\times$ is the first-order effect of an orientation error on the rotated specific force, and the orientation block $\exp(-[\boldsymbol{\omega}]_\times \Delta t)$ is the exact rotation integral (Appendix A) rather than its first-order truncation [2].

Process noise. $\mathbf{P} \leftarrow \mathbf{F}\mathbf{P}\mathbf{F}^\top + \mathbf{Q}$ with

$$\mathbf{Q} = \text{diag}(\mathbf{0}, \sigma_a^2 \Delta t^2 \mathbf{I}, \sigma_g^2 \Delta t^2 \mathbf{I}, \sigma_{ab}^2 \Delta t \mathbf{I}, \sigma_{\omega b}^2 \Delta t \mathbf{I}). \quad (6)$$

The IMU white-noise terms scale as Δt^2 : a per-sample noise of standard deviation σ perturbs the integrated velocity or orientation by $\sigma\Delta t$ over the step, injecting a variance $\sigma^2\Delta t^2$ —also the only dimensionally consistent choice. The bias random walks scale as Δt . A mis-scaled $\mathbf{Q}$ is invisible in the RMSE but glaring in the NEES of Section 7.

5 Measurement Updates

For a measurement $\mathbf{z} = h(\mathbf{x}) + \boldsymbol{\nu}$, $\boldsymbol{\nu} \sim \mathcal{N}(\mathbf{0}, \mathbf{R})$, the Jacobian $\mathbf{H} = \partial h / \partial \delta\mathbf{x}$ is evaluated at the nominal state and applied through Algorithm 1. The ten models, with the non-zero blocks of $\mathbf{H}$, are:

- **GPS position:** $h = \mathbf{p}$; $\partial h / \partial \delta\mathbf{p} = \mathbf{I}_3$.
- **GPS Doppler velocity:** $h = \mathbf{v}$; $\partial h / \partial \delta\mathbf{v} = \mathbf{I}_3$.
- **Barometer** (altitude): $h = \mathbf{e}_3^\top \mathbf{p}$; $\partial h / \partial \delta\mathbf{p} = \mathbf{e}_3^\top$.
- **Magnetometer:** $h = \mathbf{R}^\top \mathbf{m}$ ($\mathbf{m}$ the world field); $\partial h / \partial \delta\boldsymbol{\theta} = [\mathbf{R}^\top \mathbf{m}]_\times$, i.e. heading from the full vector.

- **LiDAR altimeter:** slant range to flat ground, $h = p_z/R_{33}$ with $R_{33} = \mathbf{e}_3^\top \mathbf{R} \mathbf{e}_3$; $\partial h/\partial p_z = 1/R_{33}$ and $\partial h/\partial \delta \boldsymbol{\theta} = -\frac{p_z}{R_{33}^2}[-R_{32}, R_{31}, 0]$, so a single beam senses height *and* tilt.
- **UWB radio ranging:** $h = \|\mathbf{p} - \mathbf{b}_k\|$ to beacon $\mathbf{b}_k$; $\partial h/\partial \delta \mathbf{p} = \hat{\mathbf{u}}^\top$, the unit line of sight. Position-only, so a few beacons localize with no GPS [14].
- **Optical flow:** $h = \Pi \mathbf{R}^\top \mathbf{v}$ (Π the top two rows); $\partial h/\partial \delta \mathbf{v} = \Pi \mathbf{R}^\top$, $\partial h/\partial \delta \boldsymbol{\theta} = \Pi[\mathbf{R}^\top \mathbf{v}]_\times$ [15].
- **Doppler velocity (DVL):** the full body-frame velocity $h = \mathbf{R}^\top \mathbf{v}$; $\partial h/\partial \delta \mathbf{v} = \mathbf{R}^\top$, $\partial h/\partial \delta \boldsymbol{\theta} = [\mathbf{R}^\top \mathbf{v}]_\times$.
- **Attitude fix** (star tracker / vision): a measured orientation $\mathbf{z}_q$; the innovation is $\log(\mathbf{q}^{-1} \otimes \mathbf{z}_q)$ and $\partial h/\partial \delta \boldsymbol{\theta} = \mathbf{I}_3$ —the standard multiplicative attitude update.

Every Jacobian is checked in the test suite against a central finite difference across all 15 error directions, agreeing to 10^{-4} ; the attitude and velocity updates are additionally verified to converge from a seeded error, which fails if a sign is wrong. All updates use the Joseph-form covariance step of Algorithm 1, which stays symmetric positive-definite under finite precision.

6 Injection and Reset

The correction is folded into the nominal state and the error reset to zero:

$$\mathbf{p} \leftarrow \mathbf{p} + \delta \mathbf{p}, \quad \mathbf{v} \leftarrow \mathbf{v} + \delta \mathbf{v}, \quad \mathbf{q} \leftarrow \mathbf{q} \otimes \exp\left(\frac{1}{2}\delta \boldsymbol{\theta}\right), \quad \mathbf{a}_b \leftarrow \mathbf{a}_b + \delta \mathbf{a}_b, \quad \boldsymbol{\omega}_b \leftarrow \boldsymbol{\omega}_b + \delta \boldsymbol{\omega}_b. \quad (7)$$

Because the linearization point of the orientation error moves onto the new nominal quaternion, the covariance is transformed by the reset Jacobian $\mathbf{G} = \text{diag}(\mathbf{I}, \mathbf{I}, \mathbf{I} - \frac{1}{2}[\delta \boldsymbol{\theta}]_\times, \mathbf{I}, \mathbf{I})$, $\mathbf{P} \leftarrow \mathbf{G} \mathbf{P} \mathbf{G}^\top$. Retaining the $\mathbf{I} - \frac{1}{2}[\delta \boldsymbol{\theta}]_\times$ term—often dropped—is one of the details the consistency test rewards.

7 Consistency: The NEES Gate

The decisive test is *statistical consistency* [5]: the normalized estimation error squared $\varepsilon = \mathbf{e}^\top \mathbf{P}^{-1} \mathbf{e}$, where $\mathbf{e}$ is the estimation error expressed in the filter’s own 15-dimensional tangent space (position and velocity differences, the body-frame logarithmic map for orientation, and bias differences). For a consistent filter, $\mathbb{E}[\varepsilon] = \text{dim}$. Averaged over N independent Monte-Carlo runs, $N\varepsilon \sim \chi^2(N \cdot \text{dim})$, giving the two-sided 95% acceptance interval

$$\varepsilon \in \left[\frac{\chi_{0.025}^2(N \cdot \text{dim})}{N}, \frac{\chi_{0.975}^2(N \cdot \text{dim})}{N} \right]. \quad (8)$$

Overconfidence lands above the band, conservatism below. The runs are seeded from a realistic initial error drawn from the initial covariance, so the test scores the filter and not an omniscient initialization; and the NEES is reported per five-second window as well as pooled, because consistency must hold throughout the flight, not only in the mean.

8 Simulation Environment and Test Data

Ground truth is an analytic banking helical trajectory—a horizontal orbit of radius 40 m and period 30 s with a superimposed vertical oscillation—so position, velocity, and acceleration are exact. Orientation follows a coordinated turn (nose along velocity, rolled into the turn by the bank angle that balances lateral acceleration against gravity), and the body angular rate

is obtained by central-differencing the quaternion path. From this truth a strapdown IMU and the nine aiding sensors are synthesized, each with configurable noise, bias walk, rate, and availability. A `record` command writes the raw IMU stream, every aiding measurement, and ground truth to CSV; a `replay` command re-runs the filter on that CSV and re-derives the RMSE and NEES, so the dataset is an independently verifiable artifact rather than a live-only result.

9 Results

Table 1 reports the Monte-Carlo consistency check over 40 flights of 30 s with all nine sensors active; both the position and full-state NEES fall inside the band of (8), and (not shown) each five-second window is independently within a $[2.0, 4.2]$ position-NEES band, so consistency holds throughout. The time-averaged RMSE is 0.049 m in position, 0.021 m s^{-1} in velocity, and 0.11° in attitude; replaying the recorded dataset reproduces 0.046 m.

Quantity	NEES mean	Expected	95% band
Position (3-DOF)	2.85	3.0	[2.29, 3.81]
Full state (15-DOF)	16.23	15.0	[13.35, 16.74]

Table 1: NEES consistency over 40 Monte-Carlo flights, nine sensors active. Both statistics lie inside the band—the reported uncertainty is neither over- nor under-confident.

Table 2 shows steady-state RMSE as the sensor suite is changed. With *all* aiding removed the filter dead-reckons on the IMU alone and the position error reaches hundreds of metres over 30 s—the quadratic error growth that accelerometer and gyroscope bias predict for an unaided strapdown system [7]. Any single position-fixing modality restores sub-metre accuracy: GPS-denied UWB ranging recovers from 16.6 m to 0.16 m, and the indoor and vision-aided suites—using no GPS at all—stay well under a metre.

Scenario	Position (m)	Velocity (m/s)	Attitude ($^\circ$)
Nominal (GPS + baro + mag)	0.47	0.24	0.99
IMU only (dead reckoning)	~ 300	25	8.3
GPS dropout	16.6	1.87	2.05
GPS denied + UWB ranging	0.16	0.10	0.61
Indoor (UWB + LiDAR + flow)	0.20	0.10	0.60
Vision-aided (attitude + DVL)	0.30	0.09	0.20
Full fusion (all 9 sensors)	0.05	0.02	0.11

Table 2: Steady-state RMSE by scenario (mean of 12 flights of 30 s each). Dead reckoning diverges; any single position fix, including UWB alone, recovers it.

10 Conclusion

We have presented a quaternion error-state Kalman filter fusing up to nine aiding sensors for inertial navigation, in the standard local-error formulation, with each of the ten measurement Jacobians derived and finite-difference-verified. The emphasis throughout is that correctness is *demonstrated*, not asserted: the full estimation loop passes a Monte-Carlo NEES consistency gate in aggregate and over time, and the filter reproduces the expected inertial-navigation behavior—unbounded dead-reckoning drift, and graceful recovery from any single position aid,

including GPS-denied ranging. A recorded dataset and a replay tool make the results reproducible, and the complete system runs in the browser.

A The SO(3) Exponential and the Discrete Transition

For a rotation vector $\phi \in \mathbb{R}^3$ with $\theta = \|\phi\|$ and $\hat{\phi} = \phi/\theta$, the group exponential is Rodrigues' formula

$$\exp([\phi]_{\times}) = \mathbf{I} + \frac{\sin \theta}{\theta} [\phi]_{\times} + \frac{1 - \cos \theta}{\theta^2} [\phi]_{\times}^2, \quad (9)$$

with the small-angle limit $\mathbf{I} + [\phi]_{\times}$; the corresponding unit quaternion is $\exp(\frac{1}{2}\phi) = (\cos \frac{\theta}{2}, \sin \frac{\theta}{2} \hat{\phi})$. The orientation block of (5) is $\exp(-[\omega]_{\times} \Delta t)$ evaluated with this formula rather than truncated, which matters at the high body rates of aggressive flight. The inverse (the logarithmic map) recovers the rotation vector from a unit quaternion $(w, \mathbf{v})$ as $2 \operatorname{atan2}(\|\mathbf{v}\|, w) \mathbf{v} / \|\mathbf{v}\|$, and defines the box-minus used to form the orientation component of the estimation error in Section 7.

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